

Anshia Muhammad Yaqoob

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TECHNICAL SKILLS

Mechanical: SolidWorks, AutoCAD, Fusion 360, Onshape, GD&T, 3D Printing, CNC Machining

Electronics: PCB Design, Soldering, Circuit Analysis, Oscilloscope, Multimeter, Power Supply

Embedded: Arduino, Raspberry Pi Pico, PID Control, Sensor Integration (IMU, Encoder, Ultrasonic), Motor Control

Software: C++, Python, Java, MATLAB, HTML, CSS

EXPERIENCE

Electrical & Systems Integration Lead - FIRST Tech Challenge Team 26256 Sep. 2024 – July 2025

- Led development of a closed-loop electromechanical system for a competition robot, integrating Arduino-based servo actuation and multi-sensor feedback (IMU, encoder, ultrasonic)
- Designed 10+ SolidWorks and AutoCAD layouts, defining mounting and clearances for 8 sensors, actuators, and structural components
- Designed and implemented sensor feedback loops using IMU, encoder, and ultrasonic sensors for autonomous navigation, improving path tracking accuracy by 20% through iterative calibration
- Implemented and tuned PID control in Java for drive and arm subsystems, reducing oscillation and improving response stability over 5+ iterations
- Developed 15 structured test logs in Excel to track calibration and firmware updates, enabling repeatable testing and systematic debugging

Robotics and Embedded Systems Lead Mentor

Sep. 2021 – June 2025

STEM1

Mississauga, ON

- Delivered 40+ hands-on robotics and embedded systems workshops for groups of up to 15 students using Arduino-based circuits and sensors, enabling development of functional electromechanical systems
- Designed 10 structured modules covering DC circuits, wiring, microcontroller I/O, and CAD-to-3D printing workflows for hands-on system design
- Mentored 23 student teams in designing and debugging embedded systems involving motor control, sensor integration, and iterative testing
- Managed 35 independent sessions, overseeing hardware setup, safety procedures, and multi-station integration activities

PROJECTS

Autonomous Line-Following Robot

- Built a Raspberry Pi-based mobile robot integrating 1 color sensor, 2 DC motors, 1 infrared sensor and 1 motor driver for autonomous line tracking
- Implemented and tuned a PID speed controller in Python over 6 iterations, improving trajectory stability during autonomous line tracking
- Designed and fabricated a SolidWorks chassis across 3 iterations with mounting for 5 integrated components including sensors, motors, and wiring
- Developed regulated 5V/3.3V power distribution and verified wiring across 4 subsystems using multimeter-based debugging

Mechanical CAD Design

- Produced 12 fully dimensioned 2D AutoCAD drawings for mechanical parts and subassemblies using GD&T principles
- Designed SolidWorks assemblies with up to 8 components and applied interference detection to verify fits, clearances, and fastener alignment
- Reviewed designs for manufacturability and serviceability, evaluating tool access, fastener reach, and component removal paths
- Modeled and machined a 4-part SolidWorks assembly in a machine shop, verifying dimensions using calipers and refining tolerances to improve manufacturability

Custom Printed Circuit Board (PCB) Fabrication & Circuit Design

- Designed and fabricated 7 custom PCBs for robotic control and LED driver applications using copper-clad etching, drilling, and through-hole soldering
- Developed 3 motor driver and digital logic circuits, designing PCB schematics and integrating boards as control units in mobile robotic systems
- Prototyped and tested 20+ circuits on breadboards, measuring voltage, current draw, and switching behavior using a multimeter and bench power supply
- Diagnosed and repaired 12 circuit faults including grounding issues, open traces, and signal instability through systematic debugging and solder rework

EDUCATION

University of Waterloo

Waterloo, ON

BASc Honours Mechatronics Engineering

Sep 2025 – Present

- **Relevant Coursework:** Calculus I & II, Linear Algebra, C++ Programming, Material Science, Circuits, Statics, Engineering Design
- **Certifications:** Machine Shop Safety, WHMIS, Standard First Aid (CPR C, AED)